

Yiran Jia

San Diego, United States

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Education

University of California San Diego

Expected graduation Aug 2027

Ph.D. in Mathematics (Statistics specialization)

La Jolla, CA

University of Washington

Graduated May 2021

B.S. Mathematics and B.S. Economics

Seattle, WA

Selected Publications & Preprints

Y. Jia. Multi-environment invariance learning with missing data. arXiv, 2026.

Y. Jia, E. Yeats, S. Mahan. Retraining emulation: A general framework for machine unlearning. JMM, 2026.

Y. Jia et al. Superiorization-inspired unrolled SART algorithm with U-Net perturbations. Physics in Medicine & Biology, 2022.

Computer Skills

Python; R; TensorFlow; PyTorch; MATLAB; Mathematica; Julia; Linux; LaTeX

Internship/Industry Experience

Amazon

Sep 2025 – Jan 2026

Applied Scientist II Intern, Selling Partner Experience

Seattle, WA

- Developed an automated LLM-based evaluation system for seller contact root cause summaries, designed to support analysis at the scale of 20M+ historical contacts and to produce auditable reasoning artifacts for efficient human-in-the-loop review.
- Built a multi-judge LLM deliberation framework and evaluator panel selection strategy across 19 LLMs from 8 providers, leveraging model heterogeneity and rubric-driven diversity to reduce single-model bias and improve robustness of final judgments.
- Implemented an agentic evaluation workflow with tool-calling, structured JSON prompts/outputs, and retrieval-augmented (RAG) exemplar selection for in-context learning, improving inter-rater consistency by 8%.
- Discovered that judge conformity is structurally induced by debate setup; leveraged this insight to refine deliberation structure and reduce convergence failures in hard, ambiguous cases, which comprised ~10% of evaluated cases.
- Introduced an evidence-grounding diagnostic and found that only ~30% of root cause summaries were fully supported by current case data, quantifying assumption reliance and informing stakeholder decisions on data-augmentation priorities.

Pacific Northwest National Laboratory

Jun 2025 – Sep 2025

National Security Internship Program, PhD Intern

Remote

- Proposed a unified machine unlearning framework (CURE) to selectively erase classes and concepts from neural networks and large language models via retraining emulation with knowledge distillation. [\[JMM 2026 Presentation\]](#)
- Developed novel unlearning objectives (Guided Hard Relabeling, Guided Soft Relabeling) and an orthogonal gradient method (GROGU) to balance target erasure with preservation of retained knowledge.

- Designed and implemented large-scale experimental pipelines in PyTorch to evaluate unlearning on CNN-based image classifiers (CIFAR-10/100, ImageNet) and autoregressive LLMs.
- Built evaluation protocols and metrics to quantify forgetting efficacy and model utility, demonstrating state-of-the-art unlearning performance across both vision and language domains.
- Conducted extensive ablations and theoretical analysis to characterize trade-offs between relevant-class retention and general knowledge preservation, informing practical unlearning system design.

Philips Healthcare

Jul 2018 – Jul 2019

Marketing Research Analyst

Bothell, WA

- Built SQL and R-based analytics pipelines integrating financial and text data (800+ accounts; 6,000+ news articles) to identify prospects and guide marketing strategy.
- Developed predictive models and Tableau dashboards to support pricing and sales planning, communicating insights to executives and improving data-driven decision-making.

Academia Research Experience

Sensitivity Analysis of Time-Varying Treatment Effect Estimation

Jun 2024 – Present

Supervised by Ronghui Xu, Department of Mathematics and Halicioglu Data Science Institute at UCSD

La Jolla, CA

- Develop a general framework to address unmeasured confounders with time-varying explanatory variables in longitudinal study setting, leveraging the unique panel data structure to handle both time-invariant and time-varying unobserved confounders.
- Investigate causal effects when key variables cannot be measured, providing guarantees on treatment effect estimation to help practitioners assess the robustness of their causal models.
- Designed an algorithm in R that combines domain expert knowledge on the explanatory power of confounding variables with machine learning, using double machine learning and cross-fitting to efficiently estimate the identifiable part and obtain the central limit distribution of the estimator.
- Created interpretable visualization tools to support empirical researchers in performing sensitivity analysis.

Multi-environment Causal Learning with Missing Data

Dec 2022 – 2024

Supervised by Jelena Bradic, Department of Mathematics and Halicioglu Data Science Institute at UCSD

La Jolla, CA

- Developed a data-driven model-agnostic method to leverage data heterogeneity across multiple environments, identifying stable causal factors and improving predictions in unseen settings, applying advanced statistical techniques without requiring prior knowledge of exogeneity or endogeneity. [\[arXiv preprint\]](#)
- Developed a novel objective for invariant risk minimization, tackling challenges posed by missing data and spurious associations through doubly robust imputation methods leveraging machine learning algorithms. Implemented and optimized solutions using numerical optimization techniques in Python to enhance performance and accuracy.
- Adapted the algorithm for fairness, ensuring equitable predictions across racial groups by integrating machine learning techniques and redefining environments to account for demographic diversity, while tackling missing data issues.

SART Algorithm with U-Net for CT Reconstruction

Sep, 2019 – Dec, 2022

Supervised by Thomas Humphries, Department of Mathematics at University of Washington

Bothell, WA

- Designed and implemented an innovative deep learning algorithm, trained on TensorFlow, that significantly improved CT image reconstruction, particularly in sparse-view and limited-angle scenarios. This breakthrough directly enhanced the accuracy and reliability of medical imaging, aiding early disease detection and treatment planning. [\[Code\]](#) [\[Published in Physics in Medicine Biology\]](#)
- Used Docker to create and manage development environments, making the deployment of algorithms easier and faster.
- Worked collaboratively with other researchers using GitHub for managing code versions and ensuring smooth collaboration on the project.
- Worked closely with interdisciplinary teams, presenting findings and collaborating with medical professionals to align technical advancements with practical applications in healthcare diagnostics.

Emulating Ray Tracing using Deep Generative Adversarial Network

May, 2020 – Dec, 2020

Supervised by Susana Serna and Dima Shlyakhtenko at the UCLA and AMD, IPAM RIPS 2020

Los Angeles, CA

- Developed RTGAN, a neural network model that generates images with ray tracing effects at significantly higher speeds than traditional render engines, specifically designed to address the computational expense of real-time applications like video games. [\[Code\]](#) [\[Joint Mathematics Meetings \(JMM\) Honor Presentation\]](#) [\[Paper\]](#)
- Utilized PyTorch for training the deep learning model, with Python as the main programming language for development. Collaborated with team members through GitHub for version control and project management.

Graduate Teaching Experience

- Math 281 B: Graduate Level Mathematical Statistics [\[TA Evaluation\]](#)
- Math 281 A: Graduate Level Mathematical Statistics [\[TA Evaluation\]](#)
- Math 189: Data Analysis and Inference [\[TA Evaluation\]](#)
- Math 185: Intro to Computational Stats [\[TA Evaluation\]](#)
- Math 183: Statistical Methods [\[TA Evaluation\]](#)
- Math 18: Linear Algebra [\[TA Evaluation\]](#)
- Math 20 A: Calculus I [\[TA Evaluation\]](#)
- Math 20 B: Calculus II [\[TA Evaluation\]](#)
- Math 20/10 C: Calculus III [\[TA Evaluation\]](#)

Awards & Honors

Sloan Research Fellowships

University of California San Diego

2021-2026

James B. Ax Fellowship

University of California San Diego

2021-2026

Institute for Pure & Applied Mathematics RIPS Scholar Program

University of California Los Angeles

August 2020

Honorable mention at Southern California Universities Conference

California Institute of Technology

August 2020

Relevant Course Work

- MATH 281 ABC: Three sequences statistics courses follow closely from the textbook *Asymptotic Statistics* by Vaart and *High-Dimension Statistics* by Wainwright. Produced a detailed YouTube video on [\[Rates of Least Square Estimator of Additive Nonparametric Models\]](#) from Martin Wainwright's work, offering in-depth insights for graduate-level math statistics.
- MATH 282 AB: Two sequences of applied statistic courses with advanced [\[R coding projects\]](#).

- DSC 245: Causal discovery and machine learning. Causal structure learning, identification of causal effect, estimation of causal effect, causal discovery and inference in presence of distribution shifts, selection bias, hidden confounders, cycles, nonlinear causal mechanisms, missing values, causal representation learning. [[Causal Discovery from Heterogeneous Data](#)]
- MATH 299: Causal Inference reading course covering *What If* by Hernan and Robins, and *Semiparametric Theory and Missing Data* by Tsiatis.
- MATH 287 D: High dimensional statistical learning. [[Inference for High Dimensional Generalized Estimating Equations](#)]
- MATH 287 B: Unsupervised learning. [[Theoretical Analysis of t-SNE for Clustering](#)]
- MATH 284: Survival Analysis. [[Sensitivity Analysis of Treatment Effect](#)]
- MATH 280 ABC: Three sequences of theoretical probability courses follow closely from the textbook *A Probability Path* written by Resnick.
- MATH 240 ABC: Three sequences of real-analysis courses follow closely from the textbook *Real Analysis: Modern Techniques and Their Applications* write by Folland. I created and hosted two Real Analysis YouTube series. [[Undergraduate Playlist](#)] [[Graduate Playlist](#)]
- MATH 270 ABC: Three sequences numerical analysis courses covering numerical linear algebra; numerical nonlinear algebra including polynomial approximation, interpolation, splines, and numeric quadrature; numerical ODE.
- MATH 289 A: Bootstrap. [[De-biased Ridge Regression in High Dimensional Statistics](#)]

Comprehensive/Qualifying Exams

- Mathematical Statistics - Ph.D. pass obtained in Oct 2022.
- Applied Statistics - Ph.D. pass obtained in Jan 2023.
- Numerical Analysis - Ph.D. pass obtained in June 2023.